

Actividad N°3

Programación Orientada a Objetos

(POO)

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Ejercicio 8.2 (pag 483)

Se requiere desarrollar un programa con interfaz gráfica de usuario que genere una ventana donde se solicite el ingreso de cinco notas obtenidas por un estudiante. El programa debe calcular y mostrar en la parte inferior de la ventana los siguientes datos:

- El promedio de notas ingresadas.
- La desviación estándar de las notas ingresadas.
- La mayor nota obtenida.
- La menor nota obtenida.

Ejercicio 1

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-
default.txt to change this license
 */

package com.mycompany.ejercicio1;

/**
 *
 * @author User
 */
public class Ejercicio1 {

    public static void main(String[] args) {
   VentanaPrincipal miVentanaPrincipal = new VentanaPrincipal();
   miVentanaPrincipal.setVisible(true);
    }
}
```

Notas

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-
default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit
this template
 */
```

```java
package com.mycompany.ejercicio1;

/**
 *
 * @author User
 */
public class Notas {
double[] listaNotas;

public Notas() {
    listaNotas = new double[5];
}

double calcularPromedio() {
    double suma = 0;
        for(int i=0; i < listaNotas.length; i++){
        suma = suma + listaNotas[i];
        }
    return (suma / listaNotas.length);
    }


double calcularDesviación() {
    double prom = calcularPromedio();
    double suma = 0;
        for(int i=0; i < listaNotas.length; i++) {

        suma= suma + Math.pow(listaNotas[i] - prom, 2 );
        }
    return Math.sqrt (suma/listaNotas.length );
    }

double calcularMenor()  {
    double menor = listaNotas[0];
        for(int i=0; i < listaNotas.length; i++) {
            if (listaNotas[i] < menor) {
            menor = listaNotas[i];
            }
        }
    return menor;
    }

double calcularMayor()  {
    double mayor = listaNotas[0];
        for(int i=0; i < listaNotas.length; i++) {
```

```
            if (listaNotas[i] > mayor) {
            mayor = listaNotas[i];
            }
        }
    return mayor;
    }
}
```

Ventana Principal

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-
default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/GUIForms/JFrame.java to
edit this template
 */
package com.mycompany.ejercicio1;

import javax.swing.JOptionPane;

/**
 *
 * @author User
 */
public class VentanaPrincipal extends javax.swing.JFrame {

    private static final java.util.logging.Logger logger =
java.util.logging.Logger.getLogger(VentanaPrincipal.class.getName());

    /**
     * Creates new form VentanaPrincipal
     */
    public VentanaPrincipal() {
        initComponents();
    }

    /**
     * This method is called from within the constructor to initialize the
form.
     * WARNING: Do NOT modify this code. The content of this method is
always
     * regenerated by the Form Editor.
     */
    @SuppressWarnings("unchecked")
```

```
    // <editor-fold defaultstate="collapsed" desc="Generated Code">//GEN-
BEGIN:initComponents
    private void initComponents() {

        nota1 = new javax.swing.JLabel();
        nota2 = new javax.swing.JLabel();
        nota3 = new javax.swing.JLabel();
        nota4 = new javax.swing.JLabel();
        nota5 = new javax.swing.JLabel();
        promedio = new javax.swing.JLabel();
        desviacion = new javax.swing.JLabel();
        mayor = new javax.swing.JLabel();
        campoNota1 = new javax.swing.JTextField();
        menor = new javax.swing.JLabel();
        campoNota4 = new javax.swing.JTextField();
        campoNota5 = new javax.swing.JTextField();
        campoNota2 = new javax.swing.JTextField();
        campoNota3 = new javax.swing.JTextField();
        calcular = new javax.swing.JButton();
        limpiar = new javax.swing.JButton();

        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        setTitle("Notas");

        nota1.setText("Nota 1:");

        nota2.setText("Nota 2:");

        nota3.setText("Nota 3:");

        nota4.setText("Nota 4:");

        nota5.setText("Nota 5:");

        promedio.setText("jLabel6");

        desviacion.setText("jLabel7");

        mayor.setText("jLabel8");

        campoNota1.addActionListener(this::campoNota1ActionPerformed);

        menor.setText("jLabel9");

        campoNota4.addActionListener(this::campoNota4ActionPerformed);
```

```
        campoNota3.addActionListener(this::campoNota3ActionPerformed);

        calcular.setText("Calcular");
        calcular.addActionListener(this::calcularActionPerformed);

        limpiar.setText("Limpiar");
        limpiar.addActionListener(this::limpiarActionPerformed);

        javax.swing.GroupLayout layout = new
javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEA
DING)
            .addGroup(layout.createSequentialGroup()
                .addGap(22, 22, 22)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.LEADING)
                    .addGroup(layout.createSequentialGroup()
                        .addGroup(layout.createParallelGroup(javax.swing.Gro
upLayout.Alignment.LEADING, false)
                            .addComponent(mayor,
javax.swing.GroupLayout.Alignment.TRAILING,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                            .addComponent(promedio,
javax.swing.GroupLayout.Alignment.TRAILING,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                            .addComponent(nota1,
javax.swing.GroupLayout.DEFAULT_SIZE, 63, Short.MAX_VALUE)
                            .addComponent(nota2,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                            .addComponent(nota3,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                            .addComponent(nota4,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                            .addComponent(nota5,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
```

```
                            .addComponent(desviacion,
javax.swing.GroupLayout.Alignment.TRAILING,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                            .addComponent(menor,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
                        .addPreferredGap(javax.swing.LayoutStyle.ComponentPl
acement.UNRELATED)
                        .addGroup(layout.createParallelGroup(javax.swing.Gro
upLayout.Alignment.LEADING)
                            .addComponent(campoNota2,
javax.swing.GroupLayout.PREFERRED_SIZE, 123,
javax.swing.GroupLayout.PREFERRED_SIZE)
                            .addComponent(campoNota1,
javax.swing.GroupLayout.PREFERRED_SIZE, 123,
javax.swing.GroupLayout.PREFERRED_SIZE)
                            .addComponent(campoNota3,
javax.swing.GroupLayout.PREFERRED_SIZE, 123,
javax.swing.GroupLayout.PREFERRED_SIZE)
                            .addComponent(campoNota5,
javax.swing.GroupLayout.PREFERRED_SIZE, 123,
javax.swing.GroupLayout.PREFERRED_SIZE)
                            .addComponent(campoNota4,
javax.swing.GroupLayout.PREFERRED_SIZE, 123,
javax.swing.GroupLayout.PREFERRED_SIZE))
                        .addContainerGap(28, Short.MAX_VALUE))
                    .addGroup(layout.createSequentialGroup()
                        .addComponent(calcular,
javax.swing.GroupLayout.PREFERRED_SIZE, 86,
javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addPreferredGap(javax.swing.LayoutStyle.ComponentPl
acement.RELATED)
                        .addComponent(limpiar,
javax.swing.GroupLayout.PREFERRED_SIZE, 84,
javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addGap(0, 0, Short.MAX_VALUE))))
        );
        layout.setVerticalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEA
DING)
            .addGroup(layout.createSequentialGroup()
                .addGap(24, 24, 24)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.LEADING, false)
```

```
                    .addComponent(campoNota1)
                    .addComponent(nota1,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.
RELATED)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.LEADING)
                    .addComponent(campoNota2,
javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(nota2,
javax.swing.GroupLayout.PREFERRED_SIZE, 22,
javax.swing.GroupLayout.PREFERRED_SIZE))
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.
RELATED)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.LEADING, false)
                    .addComponent(campoNota3)
                    .addComponent(nota3,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.
RELATED)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.LEADING, false)
                    .addComponent(campoNota4)
                    .addComponent(nota4,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.
RELATED)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.LEADING, false)
                    .addComponent(campoNota5)
                    .addComponent(nota5,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.
RELATED)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.BASELINE)
                    .addComponent(calcular)
                    .addComponent(limpiar))
```

```
                .addGap(18, 18, 18)
                .addComponent(promedio,
javax.swing.GroupLayout.PREFERRED_SIZE, 16,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.
RELATED)
                .addComponent(desviacion)
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.
RELATED)
                .addComponent(mayor)
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.
RELATED)
                .addComponent(menor)
                .addContainerGap(21, Short.MAX_VALUE))
        );

        pack();
    }// </editor-fold>//GEN-END:initComponents

    private void campoNota3ActionPerformed(java.awt.event.ActionEvent evt)
{//GEN-FIRST:event_campoNota3ActionPerformed
        // TODO add your handling code here:
    }//GEN-LAST:event_campoNota3ActionPerformed

    private void campoNota4ActionPerformed(java.awt.event.ActionEvent evt)
{//GEN-FIRST:event_campoNota4ActionPerformed
        // TODO add your handling code here:
    }//GEN-LAST:event_campoNota4ActionPerformed

    private void calcularActionPerformed(java.awt.event.ActionEvent evt)
{//GEN-FIRST:event_calcularActionPerformed
if (campoNota1.getText().trim().isEmpty()) {
    JOptionPane.showMessageDialog(this, "Debes ingresar la nota 1.");
    return;
}

if (campoNota2.getText().trim().isEmpty()) {
    JOptionPane.showMessageDialog(this, "Debes ingresar la nota2.");
    return;
}

if (campoNota3.getText().trim().isEmpty()) {
    JOptionPane.showMessageDialog(this, "Debes ingresar la nota 3.");
    return;
}
```

```
if (campoNota4.getText().trim().isEmpty()) {
    JOptionPane.showMessageDialog(this, "Debes ingresar la nota 4.");
    return;
}

if (campoNota5.getText().trim().isEmpty()) {
    JOptionPane.showMessageDialog(this, "Debes ingresar la nota 5.");
    return;
}

try {
    Notas notas = new Notas();

    notas.listaNotas[0] = Double.parseDouble(campoNota1.getText());
    notas.listaNotas[1] = Double.parseDouble(campoNota2.getText());
    notas.listaNotas[2] = Double.parseDouble(campoNota3.getText());
    notas.listaNotas[3] = Double.parseDouble(campoNota4.getText());
    notas.listaNotas[4] = Double.parseDouble(campoNota5.getText());

    notas.calcularPromedio();
    notas.calcularDesviación();

    promedio.setText("Promedio = " +
            String.format("%.2f", notas.calcularPromedio()));

    double desv = notas.calcularDesviación();
    desviacion.setText("Desviación estándar = " +
            String.format("%.2f", desv));

    mayor.setText("Valor mayor = " +
            String.valueOf(notas.calcularMayor()));

    menor.setText("Valor menor = " +
            String.valueOf(notas.calcularMenor()));


} catch (NumberFormatException e) {
    JOptionPane.showMessageDialog(this,
            "Debes ingresar valores númericos en las cinco notas.");
}

    }//GEN-LAST:event_calcularActionPerformed
```

```
    private void limpiarActionPerformed(java.awt.event.ActionEvent evt)
{//GEN-FIRST:event_limpiarActionPerformed
    campoNota1.setText("");
    campoNota2.setText("");
    campoNota3.setText("");
    campoNota4.setText("");
    campoNota5.setText("");

   desviacion.setText("");
   mayor.setText("");
   menor.setText("");
   promedio.setText("");
    }//GEN-LAST:event_limpiarActionPerformed

    private void campoNota1ActionPerformed(java.awt.event.ActionEvent evt)
{//GEN-FIRST:event_campoNota1ActionPerformed

    }//GEN-LAST:event_campoNota1ActionPerformed

    /**
     * @param args the command line arguments
     */
    public static void main(String args[]) {
        /* Set the Nimbus look and feel */
        //<editor-fold defaultstate="collapsed" desc=" Look and feel setting
code (optional) ">
        /* If Nimbus (introduced in Java SE 6) is not available, stay with
the default look and feel.
         * For details see
http://download.oracle.com/javase/tutorial/uiswing/lookandfeel/plaf.html
         */
        try {
            for (javax.swing.UIManager.LookAndFeelInfo info :
javax.swing.UIManager.getInstalledLookAndFeels()) {
                if ("Nimbus".equals(info.getName())) {
                    javax.swing.UIManager.setLookAndFeel(info.getClassName()
);
                    break;
                }
            }
        } catch (ReflectiveOperationException |
javax.swing.UnsupportedLookAndFeelException ex) {
            logger.log(java.util.logging.Level.SEVERE, null, ex);
        }
        //</editor-fold>
```

```
        /* Create and display the form */
        java.awt.EventQueue.invokeLater(() -> new
VentanaPrincipal().setVisible(true));
    }

    // Variables declaration - do not modify//GEN-BEGIN:variables
    private javax.swing.JButton calcular;
    private javax.swing.JTextField campoNota1;
    private javax.swing.JTextField campoNota2;
    private javax.swing.JTextField campoNota3;
    private javax.swing.JTextField campoNota4;
    private javax.swing.JTextField campoNota5;
    private javax.swing.JLabel desviacion;
    private javax.swing.JButton limpiar;
    private javax.swing.JLabel mayor;
    private javax.swing.JLabel menor;
    private javax.swing.JLabel nota1;
    private javax.swing.JLabel nota2;
    private javax.swing.JLabel nota3;
    private javax.swing.JLabel nota4;
    private javax.swing.JLabel nota5;
    private javax.swing.JLabel promedio;
    // End of variables declaration//GEN-END:variables
}
```

Diagrama de Clases

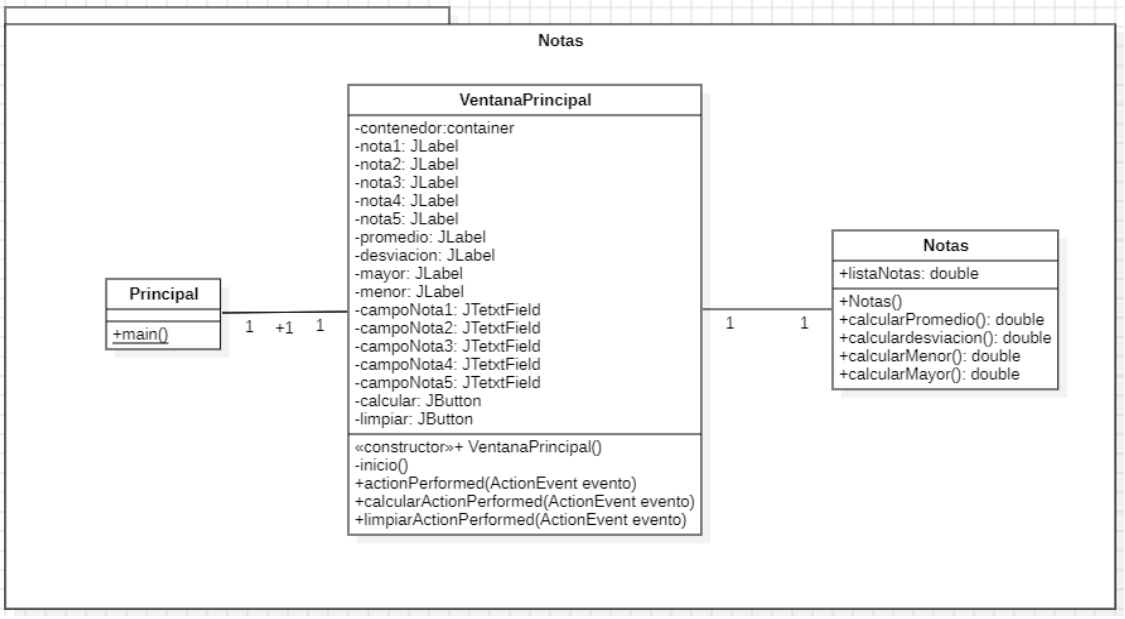


Diagrama de objetos

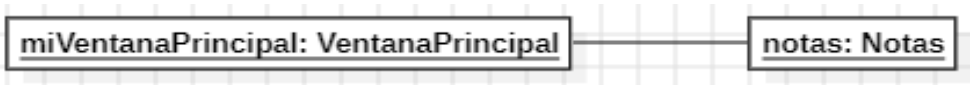
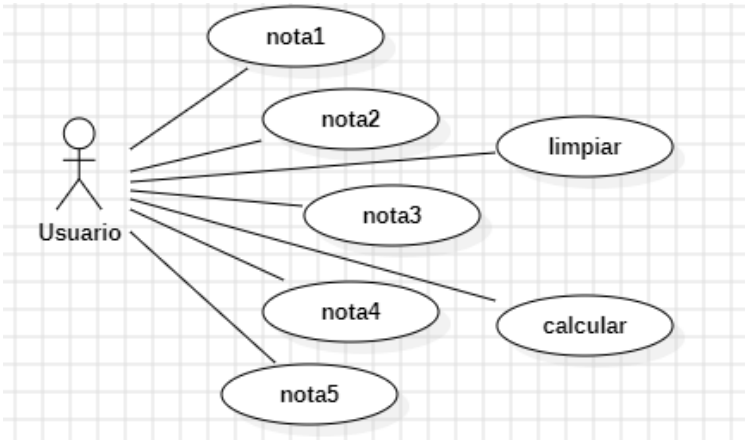


Diagrama Casos de Uso



Ejercicio 8.3 (pag 495)

Se requiere desarrollar un programa con interfaz gráfica de usuario que permita calcular el volumen y superficie de varias figuras geométricas. Las figuras geométricas son el cilindro, la esfera y la pirámide. u

- Para el cilindro se solicitan su radio y altura (en centímetros).
- Para la esfera, su radio (en centímetros).
- Para la pirámide, su base, altura y apotema (en centímetros).

Una vez ingresados estos datos, el programa calcula el volumen y superficie de cada figura. Para desarrollar el programa se debe crear una jerarquía de clases para las diferentes figuras geométricas requeridas.

Ejercicio2

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-
default.txt to change this license
 */

package com.mycompany.ejercico2;

/**
 *
 * @author User
 */
public class Ejercicio2 {

    public static void main(String[] args) {
    Ventana ventana = new Ventana();
    ventana.setVisible(true);
    }
}
```

Cilindro

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-
default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit
this template
 */
package com.mycompany.ejercico2;

/**
 *
 * @author User
 */
public class Cilindro extends FiguraGeometrica {

    private double radio;
    private double altura;

    public Cilindro(double radio, double altura) {
        this.radio = radio;
        this.altura = altura;
```

```
        volumen = Math.PI * radio * radio * altura;

        superficie = 2 * Math.PI * radio * radio
                   + 2 * Math.PI * radio * altura;
    }
}
```

VentanaCilindro

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-
default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/GUIForms/JFrame.java to
edit this template
 */
package com.mycompany.ejercico2;
import javax.swing.ImageIcon;
import java.awt.Image;
/**
 *
 * @author User
 */
public class VentanaCilindro extends javax.swing.JFrame {

    private static final java.util.logging.Logger logger =
java.util.logging.Logger.getLogger(VentanaCilindro.class.getName());

    /**
     * Creates new form VentanaCilindro
     */
    public VentanaCilindro() {
        initComponents();
    }

    /**
     * This method is called from within the constructor to initialize the
form.
     * WARNING: Do NOT modify this code. The content of this method is
always
     * regenerated by the Form Editor.
     */
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">//GEN-
BEGIN:initComponents
```

```
    private void initComponents() {

        radio = new javax.swing.JLabel();
        altura = new javax.swing.JLabel();
        volumen = new javax.swing.JLabel();
        superficie = new javax.swing.JLabel();
        CampoNota1 = new javax.swing.JTextField();
        CampoNota2 = new javax.swing.JTextField();
        calcular = new javax.swing.JButton();
        imagenCilindro = new javax.swing.JLabel();

        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        setTitle("Cilindro");

        radio.setText("Radio (cms):");

        altura.setText("Altura(cms):");

        volumen.setText("Volumen (cm3):");

        superficie.setText("Superficie (cm2):");

        calcular.setText("Calcular");
        calcular.addActionListener(this::calcularActionPerformed);

        imagenCilindro.setIcon(new
javax.swing.ImageIcon("C:\\Users\\User\\OneDrive\\Documents\\NetBeansProject
s\\Ejercico2\\src\\main\\java\\com\\mycompany\\ejercico2\\imagenes\\cilindro
.png")); // NOI18N

        javax.swing.GroupLayout layout = new
javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEA
DING)
            .addGroup(layout.createSequentialGroup()
                .addGap(20, 20, 20)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.LEADING)
                    .addGroup(layout.createSequentialGroup()
                        .addGroup(layout.createParallelGroup(javax.swing.Gro
upLayout.Alignment.LEADING)
```

```
                            .addComponent(radio,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                            .addGroup(layout.createSequentialGroup()
                                .addComponent(altura,
javax.swing.GroupLayout.PREFERRED_SIZE, 79,
javax.swing.GroupLayout.PREFERRED_SIZE)
                                .addGap(0, 57, Short.MAX_VALUE)))
                        .addPreferredGap(javax.swing.LayoutStyle.ComponentPl
acement.RELATED)
                        .addGroup(layout.createParallelGroup(javax.swing.Gro
upLayout.Alignment.LEADING)
                            .addComponent(calcular,
javax.swing.GroupLayout.PREFERRED_SIZE, 123,
javax.swing.GroupLayout.PREFERRED_SIZE)
                            .addComponent(CampoNota2,
javax.swing.GroupLayout.PREFERRED_SIZE, 123,
javax.swing.GroupLayout.PREFERRED_SIZE)
                            .addComponent(CampoNota1,
javax.swing.GroupLayout.PREFERRED_SIZE, 123,
javax.swing.GroupLayout.PREFERRED_SIZE))
                        .addGap(20, 20, 20))
                    .addGroup(layout.createSequentialGroup()
                        .addGroup(layout.createParallelGroup(javax.swing.Gro
upLayout.Alignment.LEADING)
                            .addComponent(volumen)
                            .addComponent(superficie))
                        .addPreferredGap(javax.swing.LayoutStyle.ComponentPl
acement.RELATED, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                        .addComponent(imagenCilindro,
javax.swing.GroupLayout.PREFERRED_SIZE, 69,
javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addGap(28, 28, 28))))
        );
        layout.setVerticalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEA
DING)
            .addGroup(layout.createSequentialGroup()
                .addGap(17, 17, 17)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.BASELINE)
                    .addComponent(radio,
javax.swing.GroupLayout.PREFERRED_SIZE, 22,
javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(CampoNota1))
```

```
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.
RELATED)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.BASELINE)
                    .addComponent(altura,
javax.swing.GroupLayout.PREFERRED_SIZE, 22,
javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(CampoNota2,
javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.
RELATED)
                .addComponent(calcular)
                .addGap(18, 18, 18)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.TRAILING)
                    .addGroup(layout.createSequentialGroup()
                        .addComponent(volumen)
                        .addPreferredGap(javax.swing.LayoutStyle.ComponentPl
acement.UNRELATED)
                        .addComponent(superficie))
                    .addComponent(imagenCilindro,
javax.swing.GroupLayout.PREFERRED_SIZE, 67,
javax.swing.GroupLayout.PREFERRED_SIZE))
                .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
        );

        pack();
    }// </editor-fold>//GEN-END:initComponents

    private void calcularActionPerformed(java.awt.event.ActionEvent evt)
{//GEN-FIRST:event_calcularActionPerformed

try {
    double radio = Double.parseDouble(CampoNota1.getText());
    double altura = Double.parseDouble(CampoNota2.getText());

    Cilindro cilindro = new Cilindro(radio, altura);

    volumen.setText(String.format(
        "Volumen: %.2f cm3", cilindro.getVolumen()
    ));
```

```
    superficie.setText(String.format(
        "Superficie: %.2f cm2", cilindro.getSuperficie()
    ));

} catch (NumberFormatException e) {
    javax.swing.JOptionPane.showMessageDialog(
        this,
        "Ingrese valores numericos validos."
    );
}

    }//GEN-LAST:event_calcularActionPerformed

    /**
     * @param args the command line arguments
     */
    public static void main(String args[]) {
        /* Set the Nimbus look and feel */
        //<editor-fold defaultstate="collapsed" desc=" Look and feel setting
code (optional) ">
        /* If Nimbus (introduced in Java SE 6) is not available, stay with
the default look and feel.
         * For details see
http://download.oracle.com/javase/tutorial/uiswing/lookandfeel/plaf.html
         */
        try {
            for (javax.swing.UIManager.LookAndFeelInfo info :
javax.swing.UIManager.getInstalledLookAndFeels()) {
                if ("Nimbus".equals(info.getName())) {
                    javax.swing.UIManager.setLookAndFeel(info.getClassName()
);
                    break;
                }
            }
        } catch (ReflectiveOperationException |
javax.swing.UnsupportedLookAndFeelException ex) {
            logger.log(java.util.logging.Level.SEVERE, null, ex);
        }
        //</editor-fold>

        /* Create and display the form */
        java.awt.EventQueue.invokeLater(() -> new
VentanaCilindro().setVisible(true));
    }
```

```
    // Variables declaration - do not modify//GEN-BEGIN:variables
    private javax.swing.JTextField CampoNota1;
    private javax.swing.JTextField CampoNota2;
    private javax.swing.JLabel altura;
    private javax.swing.JButton calcular;
    private javax.swing.JLabel imagenCilindro;
    private javax.swing.JLabel radio;
    private javax.swing.JLabel superficie;
    private javax.swing.JLabel volumen;
    // End of variables declaration//GEN-END:variables
}
```

Cubo

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-
default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit
this template
 */
package com.mycompany.ejercico2;

/**
 *
 * @author User
 */
public class Cubo extends FiguraGeometrica {

    private double lado;

    public Cubo(double lado) {
        this.lado = lado;

        volumen = lado * lado * lado;
        superficie = 6 * lado * lado;
    }
}
```

VentanaCubo

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-
default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/GUIForms/JFrame.java to
edit this template
 */
package com.mycompany.ejercico2;

/**
 *
 * @author User
 */
public class VentanaCubo extends javax.swing.JFrame {

    private static final java.util.logging.Logger logger =
java.util.logging.Logger.getLogger(VentanaCubo.class.getName());

    /**
     * Creates new form VentanaCubo
     */
    public VentanaCubo() {
        initComponents();
    }

    /**
     * This method is called from within the constructor to initialize the
form.
     * WARNING: Do NOT modify this code. The content of this method is
always
     * regenerated by the Form Editor.
     */
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">//GEN-
BEGIN:initComponents
    private void initComponents() {

        lado = new javax.swing.JLabel();
        CampoNota1 = new javax.swing.JTextField();
        calcular = new javax.swing.JButton();
        volumen = new javax.swing.JLabel();
        superficie = new javax.swing.JLabel();
        jLabel4 = new javax.swing.JLabel();
```

```
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        setTitle("Cubo");

        lado.setText("Lado (cm):");

        calcular.setText("Calcular");
        calcular.addActionListener(this::calcularActionPerformed);

        volumen.setText("Volumen (cm3):");

        superficie.setText("Superficie(cm2):");

        jLabel4.setIcon(new
javax.swing.ImageIcon("C:\\Users\\User\\OneDrive\\Documents\\NetBeansProject
s\\Ejercico2\\src\\main\\java\\com\\mycompany\\ejercico2\\imagenes\\cubo.png
")); // NOI18N

        javax.swing.GroupLayout layout = new
javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEA
DING)
            .addGroup(layout.createSequentialGroup()
                .addGap(28, 28, 28)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.TRAILING, false)
                    .addComponent(superficie,
javax.swing.GroupLayout.Alignment.LEADING,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                    .addComponent(volumen,
javax.swing.GroupLayout.Alignment.LEADING,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                    .addComponent(lado,
javax.swing.GroupLayout.Alignment.LEADING,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
                .addGap(18, 18, 18)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.LEADING, false)
                    .addComponent(calcular,
javax.swing.GroupLayout.DEFAULT_SIZE, 109, Short.MAX_VALUE)
                    .addComponent(CampoNota1)
```

```
                    .addComponent(jLabel4,
javax.swing.GroupLayout.Alignment.TRAILING,
javax.swing.GroupLayout.PREFERRED_SIZE, 79,
javax.swing.GroupLayout.PREFERRED_SIZE))
                .addContainerGap(18, Short.MAX_VALUE))
        );
        layout.setVerticalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEA
DING)
            .addGroup(layout.createSequentialGroup()
                .addGap(24, 24, 24)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.LEADING, false)
                    .addComponent(CampoNota1,
javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addGroup(layout.createSequentialGroup()
                        .addGap(3, 3, 3)
                        .addComponent(lado,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)))
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.LEADING)
                    .addGroup(layout.createSequentialGroup()
                        .addGap(18, 18, 18)
                        .addComponent(calcular)
                        .addPreferredGap(javax.swing.LayoutStyle.ComponentPl
acement.UNRELATED)
                        .addComponent(jLabel4))
                    .addGroup(layout.createSequentialGroup()
                        .addGap(55, 55, 55)
                        .addComponent(volumen)
                        .addPreferredGap(javax.swing.LayoutStyle.ComponentPl
acement.UNRELATED)
                        .addComponent(superficie)))
                .addContainerGap(8, Short.MAX_VALUE))
        );

        pack();
    }// </editor-fold>//GEN-END:initComponents

    private void calcularActionPerformed(java.awt.event.ActionEvent evt)
{//GEN-FIRST:event_calcularActionPerformed
```

```
try {
    double lado = Double.parseDouble(CampoNota1.getText());

    Cubo cubo = new Cubo(lado);

    volumen.setText(String.format(
        "Volumen: %.2f cm3", cubo.getVolumen()
    ));

    superficie.setText(String.format(
        "Superficie: %.2f cm2", cubo.getSuperficie()
    ));

} catch (NumberFormatException e) {
    javax.swing.JOptionPane.showMessageDialog(
        this,
        "Ingrese un valor numerico valido."
    );
}
    }//GEN-LAST:event_calcularActionPerformed

    /**
     * @param args the command line arguments
     */
    public static void main(String args[]) {
        /* Set the Nimbus look and feel */
        //<editor-fold defaultstate="collapsed" desc=" Look and feel setting
code (optional) ">
        /* If Nimbus (introduced in Java SE 6) is not available, stay with
the default look and feel.
         * For details see
http://download.oracle.com/javase/tutorial/uiswing/lookandfeel/plaf.html
         */
        try {
            for (javax.swing.UIManager.LookAndFeelInfo info :
javax.swing.UIManager.getInstalledLookAndFeels()) {
                if ("Nimbus".equals(info.getName())) {
                    javax.swing.UIManager.setLookAndFeel(info.getClassName()
);
                    break;
                }
            }
        } catch (ReflectiveOperationException |
javax.swing.UnsupportedLookAndFeelException ex) {
            logger.log(java.util.logging.Level.SEVERE, null, ex);
```

```
        }
        //</editor-fold>

        /* Create and display the form */
        java.awt.EventQueue.invokeLater(() -> new
VentanaCubo().setVisible(true));
    }

    // Variables declaration - do not modify//GEN-BEGIN:variables
    private javax.swing.JTextField CampoNota1;
    private javax.swing.JButton calcular;
    private javax.swing.JLabel jLabel4;
    private javax.swing.JLabel lado;
    private javax.swing.JLabel superficie;
    private javax.swing.JLabel volumen;
    // End of variables declaration//GEN-END:variables
}
```

Esfera

```
/*
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default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit
this template
 */
package com.mycompany.ejercico2;

/**
 *
 * @author User
 */
public class Cubo extends FiguraGeometrica {

    private double lado;

    public Cubo(double lado) {
        this.lado = lado;

        volumen = lado * lado * lado;
        superficie = 6 * lado * lado;
    }
}
```

VentanaEsfera

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-
default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/GUIForms/JFrame.java to
edit this template
 */
package com.mycompany.ejercico2;

/**
 *
 * @author User
 */
public class VentanaEsfera extends javax.swing.JFrame {

    private static final java.util.logging.Logger logger =
java.util.logging.Logger.getLogger(VentanaEsfera.class.getName());

    /**
     * Creates new form VentanaEsfera
     */
    public VentanaEsfera() {
        initComponents();
    }

    /**
     * This method is called from within the constructor to initialize the
form.
     * WARNING: Do NOT modify this code. The content of this method is
always
     * regenerated by the Form Editor.
     */
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">//GEN-
BEGIN:initComponents
    private void initComponents() {

        calcular = new javax.swing.JButton();
        radio = new javax.swing.JLabel();
        volumen = new javax.swing.JLabel();
        superficie = new javax.swing.JLabel();
        CampoNota1 = new javax.swing.JTextField();
        jLabel2 = new javax.swing.JLabel();
```

```
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        setTitle("Esfera");

        calcular.setText("Calcular");
        calcular.addActionListener(this::calcularActionPerformed);

        radio.setText("Radio (cms):");

        volumen.setText("Volumen (cm3):");

        superficie.setText("Superficie (cm2):");

        jLabel2.setIcon(new
javax.swing.ImageIcon("C:\\Users\\User\\OneDrive\\Documents\\NetBeansProject
s\\Ejercico2\\src\\main\\java\\com\\mycompany\\ejercico2\\imagenes\\esfera.p
ng")); // NOI18N

        javax.swing.GroupLayout layout = new
javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEA
DING)
            .addGroup(layout.createSequentialGroup()
                .addContainerGap()
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.TRAILING, false)
                    .addComponent(superficie,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                    .addGroup(layout.createParallelGroup(javax.swing.GroupLa
yout.Alignment.TRAILING, false)
                        .addComponent(volumen,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                        .addComponent(radio,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)))
                .addGap(18, 18, 18)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.LEADING, false)
                    .addComponent(CampoNota1)
                    .addComponent(calcular,
javax.swing.GroupLayout.DEFAULT_SIZE, 119, Short.MAX_VALUE)
```

```
                    .addGroup(javax.swing.GroupLayout.Alignment.TRAILING,
layout.createSequentialGroup()
                        .addComponent(jLabel2,
javax.swing.GroupLayout.PREFERRED_SIZE, 78,
javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addGap(19, 19, 19)))
                .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
        );
        layout.setVerticalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEA
DING)
            .addGroup(javax.swing.GroupLayout.Alignment.TRAILING,
layout.createSequentialGroup()
                .addGap(20, 20, 20)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.LEADING, false)
                    .addComponent(CampoNota1,
javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addGroup(layout.createSequentialGroup()
                        .addGap(3, 3, 3)
                        .addComponent(radio,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)))
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.
UNRELATED)
                .addComponent(calcular)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.LEADING)
                    .addGroup(layout.createSequentialGroup()
                        .addPreferredGap(javax.swing.LayoutStyle.ComponentPl
acement.RELATED, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                        .addComponent(volumen)
                        .addPreferredGap(javax.swing.LayoutStyle.ComponentPl
acement.RELATED)
                        .addComponent(superficie)
                        .addGap(15, 15, 15))
                    .addGroup(layout.createSequentialGroup()
                        .addGap(18, 18, 18)
                        .addComponent(jLabel2,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                        .addContainerGap())))
```

```
        );

        pack();
    }// </editor-fold>//GEN-END:initComponents

    private void calcularActionPerformed(java.awt.event.ActionEvent evt)
{//GEN-FIRST:event_calcularActionPerformed

try {
    double radio = Double.parseDouble(CampoNota1.getText());

    Esfera esfera = new Esfera(radio);

    volumen.setText(String.format(
        "Volumen: %.2f cm3", esfera.getVolumen()
    ));

    superficie.setText(String.format(
        "Superficie: %.2f cm2", esfera.getSuperficie()
    ));

} catch (NumberFormatException e) {
    javax.swing.JOptionPane.showMessageDialog(
        this,
        "Ingrese un valor numerico valido."
    );
}

    }//GEN-LAST:event_calcularActionPerformed

    /**
     * @param args the command line arguments
     */
    public static void main(String args[]) {
        /* Set the Nimbus look and feel */
        //<editor-fold defaultstate="collapsed" desc=" Look and feel setting
code (optional) ">
        /* If Nimbus (introduced in Java SE 6) is not available, stay with
the default look and feel.
         * For details see
http://download.oracle.com/javase/tutorial/uiswing/lookandfeel/plaf.html
         */
        try {
            for (javax.swing.UIManager.LookAndFeelInfo info :
javax.swing.UIManager.getInstalledLookAndFeels()) {
```

```
                if ("Nimbus".equals(info.getName())) {
                    javax.swing.UIManager.setLookAndFeel(info.getClassName()
);
                    break;
                }
            }
        } catch (ReflectiveOperationException |
javax.swing.UnsupportedLookAndFeelException ex) {
            logger.log(java.util.logging.Level.SEVERE, null, ex);
        }
        //</editor-fold>

        /* Create and display the form */
        java.awt.EventQueue.invokeLater(() -> new
VentanaEsfera().setVisible(true));
    }

    // Variables declaration - do not modify//GEN-BEGIN:variables
    private javax.swing.JTextField CampoNota1;
    private javax.swing.JButton calcular;
    private javax.swing.JLabel jLabel2;
    private javax.swing.JLabel radio;
    private javax.swing.JLabel superficie;
    private javax.swing.JLabel volumen;
    // End of variables declaration//GEN-END:variables
}
```

Piramide

```
/*
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default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit
this template
 */
package com.mycompany.ejercico2;

/**
 *
 * @author User
 */
public class Piramide extends FiguraGeometrica {
    private double base;
```

```
    private double altura;
    private double apotema;

    public Piramide(double base, double altura,
                    double apotema) {
        this.base = base;
        this.altura = altura;
        this.apotema = apotema;

        volumen = (base * base * altura) / 3;

        superficie = base * base + 2 * base * apotema;
    }
}
```

Ventanapiramide

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-
default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/GUIForms/JFrame.java to
edit this template
 */
package com.mycompany.ejercico2;

/**
 *
 * @author User
 */
public class VentanaPiramide extends javax.swing.JFrame {

    private static final java.util.logging.Logger logger =
java.util.logging.Logger.getLogger(VentanaPiramide.class.getName());

    /**
     * Creates new form VentanaPiramide
     */
    public VentanaPiramide() {
        initComponents();
    }

    /**
```

```
     * This method is called from within the constructor to initialize the
form.
     * WARNING: Do NOT modify this code. The content of this method is
always
     * regenerated by the Form Editor.
     */
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">//GEN-
BEGIN:initComponents
    private void initComponents() {

        base = new javax.swing.JLabel();
        altura = new javax.swing.JLabel();
        apotema = new javax.swing.JLabel();
        volumen = new javax.swing.JLabel();
        superficie = new javax.swing.JLabel();
        calcular = new javax.swing.JButton();
        CampoNota1 = new javax.swing.JTextField();
        CampoNota2 = new javax.swing.JTextField();
        CampoNota3 = new javax.swing.JTextField();
        jLabel2 = new javax.swing.JLabel();

        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        setTitle("Piramide");

        base.setText("Base (cms):");

        altura.setText("Altura (cms):");

        apotema.setText("Apotema (cms):");

        volumen.setText("Volumen (cm3):");

        superficie.setText("Superficie (cm2):");

        calcular.setText("Calcular");
        calcular.addActionListener(this::calcularActionPerformed);

        jLabel2.setIcon(new
javax.swing.ImageIcon("C:\\Users\\User\\OneDrive\\Documents\\NetBeansProject
s\\Ejercico2\\src\\main\\java\\com\\mycompany\\ejercico2\\imagenes\\piramide
.png")); // NOI18N

        javax.swing.GroupLayout layout = new
javax.swing.GroupLayout(getContentPane());
```

```
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEA
DING)
            .addGroup(layout.createSequentialGroup()
                .addGap(25, 25, 25)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.LEADING)
                    .addGroup(layout.createParallelGroup(javax.swing.GroupLa
yout.Alignment.TRAILING, false)
                        .addComponent(superficie,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                        .addGroup(layout.createParallelGroup(javax.swing.Gro
upLayout.Alignment.TRAILING, false)
                            .addComponent(base,
javax.swing.GroupLayout.Alignment.LEADING,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                            .addComponent(altura,
javax.swing.GroupLayout.Alignment.LEADING,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                            .addComponent(apotema,
javax.swing.GroupLayout.Alignment.LEADING,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)))
                    .addComponent(volumen))
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.
RELATED, 84, Short.MAX_VALUE)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.LEADING)
                    .addGroup(layout.createParallelGroup(javax.swing.GroupLa
yout.Alignment.LEADING, false)
                        .addComponent(calcular,
javax.swing.GroupLayout.DEFAULT_SIZE, 109, Short.MAX_VALUE)
                        .addComponent(CampoNota1,
javax.swing.GroupLayout.Alignment.TRAILING)
                        .addComponent(CampoNota2)
                        .addComponent(CampoNota3))
                    .addComponent(jLabel2))
                .addContainerGap(12, Short.MAX_VALUE))
        );
        layout.setVerticalGroup(
```

```
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEA
DING)
            .addGroup(layout.createSequentialGroup()
                .addGap(21, 21, 21)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.BASELINE)
                    .addComponent(base)
                    .addComponent(CampoNota1,
javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.
UNRELATED)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.BASELINE)
                    .addComponent(altura)
                    .addComponent(CampoNota2,
javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
                .addGap(10, 10, 10)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.BASELINE)
                    .addComponent(CampoNota3,
javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(apotema))
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.
UNRELATED)
                .addComponent(calcular)
                .addGap(18, 18, 18)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.LEADING)
                    .addGroup(layout.createSequentialGroup()
                        .addComponent(volumen)
                        .addPreferredGap(javax.swing.LayoutStyle.ComponentPl
acement.RELATED)
                        .addComponent(superficie)
                        .addGap(0, 0, Short.MAX_VALUE))
                    .addComponent(jLabel2,
javax.swing.GroupLayout.DEFAULT_SIZE, 86, Short.MAX_VALUE))
                .addContainerGap())
        );
```

```
        pack();
    }// </editor-fold>//GEN-END:initComponents

    private void calcularActionPerformed(java.awt.event.ActionEvent evt)
{//GEN-FIRST:event_calcularActionPerformed

try {
    double base = Double.parseDouble(CampoNota1.getText());
    double altura = Double.parseDouble(CampoNota2.getText());
    double apotema = Double.parseDouble(CampoNota3.getText());

    Piramide piramide = new Piramide(base, altura, apotema);

    volumen.setText(String.format(
        "Volumen: %.2f cm3", piramide.getVolumen()
    ));

    superficie.setText(String.format(
        "Superficie: %.2f cm2", piramide.getSuperficie()
    ));

} catch (NumberFormatException e) {
    javax.swing.JOptionPane.showMessageDialog(
        this,
        "Ingrese valores numericos validos."
    );
}
    }//GEN-LAST:event_calcularActionPerformed

    /**
     * @param args the command line arguments
     */
    public static void main(String args[]) {
        /* Set the Nimbus look and feel */
        //<editor-fold defaultstate="collapsed" desc=" Look and feel setting
code (optional) ">
        /* If Nimbus (introduced in Java SE 6) is not available, stay with
the default look and feel.
         * For details see
http://download.oracle.com/javase/tutorial/uiswing/lookandfeel/plaf.html
         */
        try {
            for (javax.swing.UIManager.LookAndFeelInfo info :
javax.swing.UIManager.getInstalledLookAndFeels()) {
                if ("Nimbus".equals(info.getName())) {
```

```
                    javax.swing.UIManager.setLookAndFeel(info.getClassName()
);
                    break;
                }
            }
        } catch (ReflectiveOperationException |
javax.swing.UnsupportedLookAndFeelException ex) {
            logger.log(java.util.logging.Level.SEVERE, null, ex);
        }
        //</editor-fold>

        /* Create and display the form */
        java.awt.EventQueue.invokeLater(() -> new
VentanaPiramide().setVisible(true));
    }

    // Variables declaration - do not modify//GEN-BEGIN:variables
    private javax.swing.JTextField CampoNota1;
    private javax.swing.JTextField CampoNota2;
    private javax.swing.JTextField CampoNota3;
    private javax.swing.JLabel altura;
    private javax.swing.JLabel apotema;
    private javax.swing.JLabel base;
    private javax.swing.JButton calcular;
    private javax.swing.JLabel jLabel2;
    private javax.swing.JLabel superficie;
    private javax.swing.JLabel volumen;
    // End of variables declaration//GEN-END:variables
}
```

Prisma

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-
default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit
this template
 */
package com.mycompany.ejercico2;

/**
 *
 * @author User
```

```
 */
public class Prisma extends FiguraGeometrica {

    private double largo;
    private double ancho;
    private double altura;

    public Prisma(double largo, double ancho, double altura) {
        this.largo = largo;
        this.ancho = ancho;
        this.altura = altura;

        volumen = largo * ancho * altura;

        superficie = 2 * (largo * ancho
                + largo * altura
                + ancho * altura);
    }
}
```

VentanaPrisma

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-
default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/GUIForms/JFrame.java to
edit this template
 */
package com.mycompany.ejercico2;

/**
 *
 * @author User
 */
public class VentanaPrisma extends javax.swing.JFrame {

    private static final java.util.logging.Logger logger =
java.util.logging.Logger.getLogger(VentanaPrisma.class.getName());

    /**
     * Creates new form VentanaPrisma
     */
    public VentanaPrisma() {
        initComponents();
```

```
    }

    /**
     * This method is called from within the constructor to initialize the
form.
     * WARNING: Do NOT modify this code. The content of this method is
always
     * regenerated by the Form Editor.
     */
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">//GEN-
BEGIN:initComponents
    private void initComponents() {

        largo = new javax.swing.JLabel();
        ancho = new javax.swing.JLabel();
        altura = new javax.swing.JLabel();
        volumen = new javax.swing.JLabel();
        superficie = new javax.swing.JLabel();
        CampoNota1 = new javax.swing.JTextField();
        CampoNota2 = new javax.swing.JTextField();
        CampoNota3 = new javax.swing.JTextField();
        calcular = new javax.swing.JButton();
        jLabel1 = new javax.swing.JLabel();

        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        setTitle("Prisma");

        largo.setText("Largo (cm):");

        ancho.setText("Ancho (cm):");

        altura.setText("Altura (cm):");

        volumen.setText("Volumen (cms3):");

        superficie.setText("Superficie (cms2)");

        CampoNota2.addActionListener(this::CampoNota2ActionPerformed);

        calcular.setText("Calcular");
        calcular.addActionListener(this::calcularActionPerformed);

        jLabel1.setIcon(new
javax.swing.ImageIcon("C:\\Users\\User\\OneDrive\\Documents\\NetBeansProject
```

```
s\\Ejercico2\\src\\main\\java\\com\\mycompany\\ejercico2\\imagenes\\prisma.p
ng")); // NOI18N

        javax.swing.GroupLayout layout = new
javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEA
DING)
            .addGroup(layout.createSequentialGroup()
                .addContainerGap()
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.TRAILING)
                    .addGroup(layout.createParallelGroup(javax.swing.GroupLa
yout.Alignment.LEADING)
                        .addGroup(layout.createParallelGroup(javax.swing.Gro
upLayout.Alignment.TRAILING)
                            .addComponent(altura,
javax.swing.GroupLayout.PREFERRED_SIZE, 76,
javax.swing.GroupLayout.PREFERRED_SIZE)
                            .addComponent(ancho,
javax.swing.GroupLayout.PREFERRED_SIZE, 76,
javax.swing.GroupLayout.PREFERRED_SIZE)
                            .addComponent(largo,
javax.swing.GroupLayout.PREFERRED_SIZE, 76,
javax.swing.GroupLayout.PREFERRED_SIZE))
                        .addGroup(layout.createParallelGroup(javax.swing.Gro
upLayout.Alignment.TRAILING)
                            .addComponent(calcular,
javax.swing.GroupLayout.PREFERRED_SIZE, 109,
javax.swing.GroupLayout.PREFERRED_SIZE)
                            .addGroup(javax.swing.GroupLayout.Alignment.LEAD
ING, layout.createSequentialGroup()
                                .addGap(78, 78, 78)
                                .addGroup(layout.createParallelGroup(javax.s
wing.GroupLayout.Alignment.LEADING, false)
                                    .addComponent(CampoNota3,
javax.swing.GroupLayout.DEFAULT_SIZE, 109, Short.MAX_VALUE)
                                    .addComponent(CampoNota2)
                                    .addComponent(CampoNota1)))))
                    .addGroup(layout.createSequentialGroup()
                        .addGroup(layout.createParallelGroup(javax.swing.Gro
upLayout.Alignment.LEADING)
                            .addComponent(volumen)
```

```
                            .addComponent(superficie,
javax.swing.GroupLayout.PREFERRED_SIZE, 100,
javax.swing.GroupLayout.PREFERRED_SIZE))
                        .addGap(18, 18, 18)
                        .addComponent(jLabel1,
javax.swing.GroupLayout.PREFERRED_SIZE, 79,
javax.swing.GroupLayout.PREFERRED_SIZE)))
                .addContainerGap(28, Short.MAX_VALUE))
        );
        layout.setVerticalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEA
DING)
            .addGroup(layout.createSequentialGroup()
                .addGap(17, 17, 17)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.BASELINE)
                    .addComponent(largo,
javax.swing.GroupLayout.PREFERRED_SIZE, 19,
javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(CampoNota1,
javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
                .addGap(6, 6, 6)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.BASELINE)
                    .addComponent(ancho,
javax.swing.GroupLayout.PREFERRED_SIZE, 19,
javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(CampoNota2,
javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.
RELATED)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.BASELINE)
                    .addComponent(altura,
javax.swing.GroupLayout.PREFERRED_SIZE, 22,
javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(CampoNota3,
javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
```

```
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.
RELATED)
                .addComponent(calcular)
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.
UNRELATED)
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout
.Alignment.LEADING)
                    .addGroup(layout.createSequentialGroup()
                        .addComponent(volumen,
javax.swing.GroupLayout.PREFERRED_SIZE, 22,
javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addPreferredGap(javax.swing.LayoutStyle.ComponentPl
acement.RELATED)
                        .addComponent(superficie,
javax.swing.GroupLayout.PREFERRED_SIZE, 25,
javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addGap(0, 0, Short.MAX_VALUE))
                    .addGroup(layout.createSequentialGroup()
                        .addGap(3, 3, 3)
                        .addComponent(jLabel1,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)))
                .addContainerGap())
        );

        pack();
    }// </editor-fold>//GEN-END:initComponents

    private void CampoNota2ActionPerformed(java.awt.event.ActionEvent evt)
{//GEN-FIRST:event_CampoNota2ActionPerformed
        // TODO add your handling code here:
    }//GEN-LAST:event_CampoNota2ActionPerformed

    private void calcularActionPerformed(java.awt.event.ActionEvent evt)
{//GEN-FIRST:event_calcularActionPerformed

try {
    double largo = Double.parseDouble(CampoNota1.getText());
    double ancho = Double.parseDouble(CampoNota2.getText());
    double altura = Double.parseDouble(CampoNota3.getText());

    Prisma prisma = new Prisma(largo, ancho, altura);

    volumen.setText(String.format(
        "Volumen: %.2f cm3", prisma.getVolumen()
```

```
    ));

    superficie.setText(String.format(
        "Superficie: %.2f cm2", prisma.getSuperficie()
    ));

} catch (NumberFormatException e) {
    javax.swing.JOptionPane.showMessageDialog(
        this,
        "Ingrese valores numericos validos."
    );
}
    }//GEN-LAST:event_calcularActionPerformed

    /**
     * @param args the command line arguments
     */
    public static void main(String args[]) {
        /* Set the Nimbus look and feel */
        //<editor-fold defaultstate="collapsed" desc=" Look and feel setting
code (optional) ">
        /* If Nimbus (introduced in Java SE 6) is not available, stay with
the default look and feel.
         * For details see
http://download.oracle.com/javase/tutorial/uiswing/lookandfeel/plaf.html
         */
        try {
            for (javax.swing.UIManager.LookAndFeelInfo info :
javax.swing.UIManager.getInstalledLookAndFeels()) {
                if ("Nimbus".equals(info.getName())) {
                    javax.swing.UIManager.setLookAndFeel(info.getClassName()
);
                    break;
                }
            }
        } catch (ReflectiveOperationException |
javax.swing.UnsupportedLookAndFeelException ex) {
            logger.log(java.util.logging.Level.SEVERE, null, ex);
        }
        //</editor-fold>

        /* Create and display the form */
        java.awt.EventQueue.invokeLater(() -> new
VentanaPrisma().setVisible(true));
    }
```

```
    // Variables declaration - do not modify//GEN-BEGIN:variables
    private javax.swing.JTextField CampoNota1;
    private javax.swing.JTextField CampoNota2;
    private javax.swing.JTextField CampoNota3;
    private javax.swing.JLabel altura;
    private javax.swing.JLabel ancho;
    private javax.swing.JButton calcular;
    private javax.swing.JLabel jLabel1;
    private javax.swing.JLabel largo;
    private javax.swing.JLabel superficie;
    private javax.swing.JLabel volumen;
    // End of variables declaration//GEN-END:variables
}
```

Diagrama de Clases

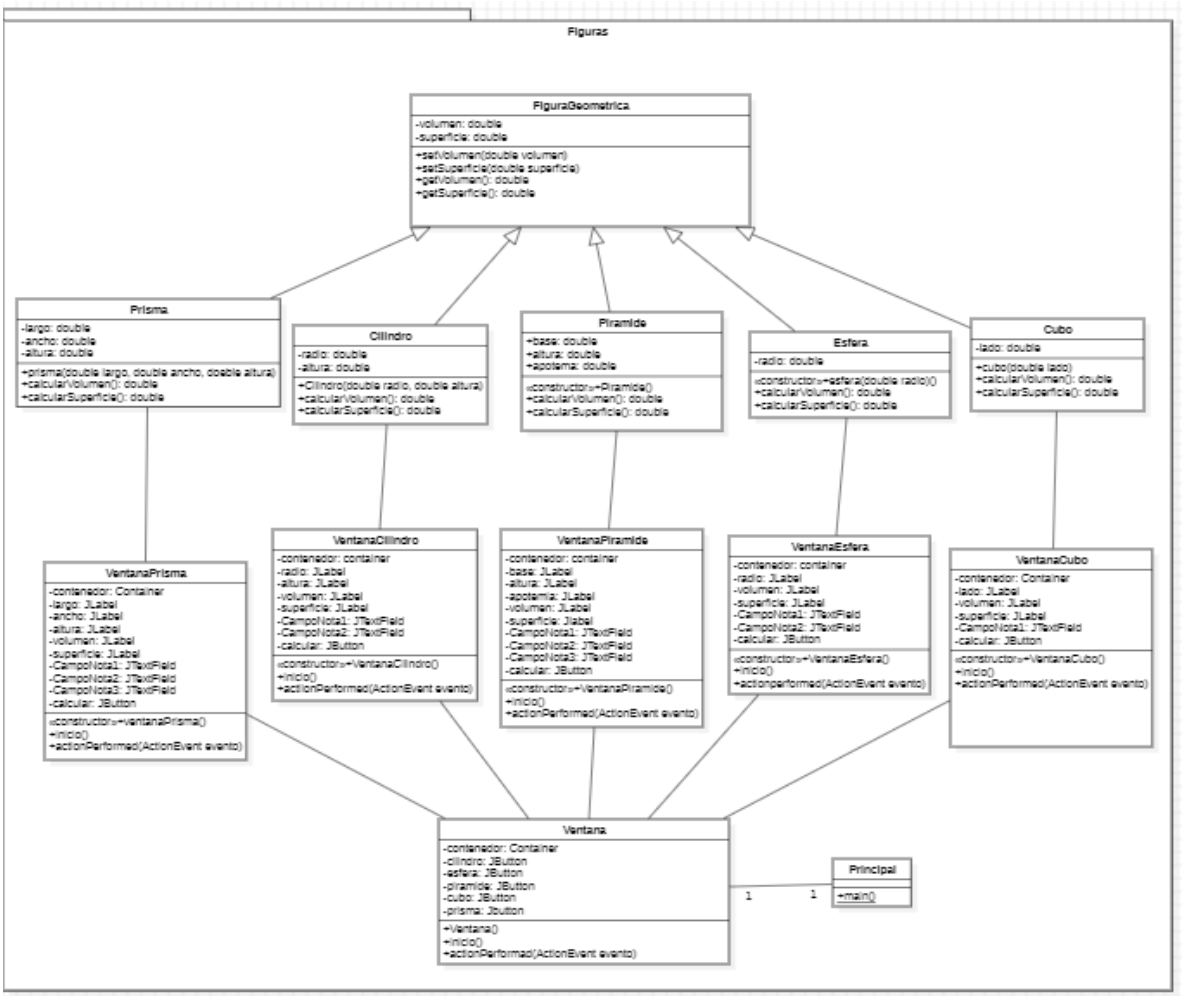


Diagrama de Objetos

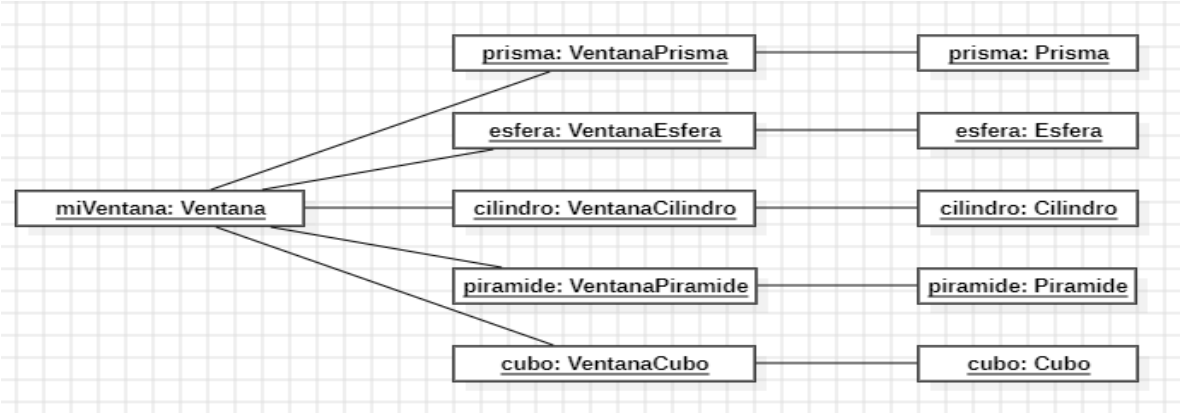


Diagrama Casos de Uso

